

CSCI 1300 ASSIGNMENT 3 – RESPONSIVE REDESIGN

PART 1: IDENTIFYING USABILITY PROBLEMS FOR PENNY JUICE WEBSITE

(<https://www.pennyjuice.com/>)



Rationale:

I chose the Penny Juice website because it's a business-oriented site with clear purpose (selling concentrated juice to childcare centers) but suffers from significant usability issues despite its simple goal. The outdated design and cluttered interface provide numerous opportunities for meaningful improvement.

Usability Problems:

LEARNABILITY	MEMORABILITY	EFFICIENCY	CONCEPTUAL MODEL ISSUES
• Animated GIFs, flashing elements, and clashing colors create a confusing homepage experience. • Key information is buried beneath decorative elements. • Overlapping, illogically sequenced content categories confuse site navigation.	• Inconsistent page structures force users to relearn the interface repeatedly. • Lack of visual indicators for visited sections or completed actions. • Related information is scattered across pages without logical connections, hindering information retrieval.	• No direct search functionality, forcing menu navigation for specific information. • Order forms have confusing layouts and require unnecessary information.	• Childlike aesthetic clashes with B2B e-commerce purpose for childcare professionals. • Navigation structure fails to match how childcare professionals purchase juice concentrate.

WebAIM WAVE Results:

- **Errors:** Multiple images missing alternative text, making content inaccessible to screen readers
- **Contrast Issues:** Poor color contrast between text and background in multiple sections, making content difficult to read
- **Missing Form Labels:** Form fields lack proper labels, creating barriers for users with screen readers
- **Empty Links:** Several links have no descriptive text
- **Deprecated HTML:** Use of outdated HTML practices that affect assistive technology compatibility
- **Missing Document Structure:** Lack of proper heading hierarchy and structural elements

I agree with the WAVE findings. The website has significant accessibility issues that would prevent users with disabilities from effectively navigating and using the site. The reliance on visual elements without proper text alternatives is particularly problematic for visually impaired users, while the inconsistent navigation and poor structure would create difficulties for anyone using keyboard navigation or screen readers.

PART 2: VISUAL REDESIGN

Visual Design Style Guide:

01. LOGOS	02. BUTTONS	03. COLOR PALETTE
 PENNY JUICE PENNY JUICE	ORDER NOW VIEW PRODUCTS LEARN MORE CONTACT US Next Next	Primary Palette Brand Blue Hex #1E88E5 Brand Yellow Hex #FFD54F Brand Green Hex #43A047 Text Gray Hex #424242 Secondary Palette Raspberry Hex #E91E63 Light Blue Hex #90CAF9 Orange Hex #FF9800 White Hex #FFFFFF
04. TEXT STYLES	05. ICONS	06. CONTROLS
Montserrat Aa Aa Aa Bold Medium Regular Montserrat Bold 24px Montserrat Bold 18px Montserrat Bold 14px Open Sans Regular 16px Open Sans Regular 14px Nunito Regular 16px (Accent)	 Product  Benefits  Order  Default  Hover  Active UI STATES FOLLOW CONSISTENT PATTERNS Hover: Lighten by 10% Active: Darken by 10% Focus: Blue outline	Toggle Add & Subtract Buttons - + - + Check Boxes RESPONSIVE DESIGN PRINCIPLES All elements use rem units and flex/grid layouts

Desktop View Mockup:

Desktop View (1920px)

Horizontal nav on desktop - display: flex; justify-content: space-between

PENNY JUICE

HOME

PRODUCTS

BENEFITS

NUTRITION

ABOUT

ORDER NOW

Healthy Hydration
for Happy Kids

100% natural juice concentrate for childcare centers and schools. No added sugars, colors, or preservatives.

ORDER ONLINE

VIEW PRODUCTS

100% Natural Juice Concentrate!

Our Products

Citrus Blend

Orange, lemon, and lime

LEARN MORE

Berry Burst

Strawberry, blueberry, raspberry

LEARN MORE

Tropical Mix

Pineapple, mango, passion fruit

LEARN MORE

Benefits

100%

All Natural

No artificial flavors, colors, or preservatives!

\$

Cost Effective

Save up to 40% compared to pre-packaged juice

Vit

Nutritious

Packed with essential vitamins and minerals

★

Kid Approved

Delicious flavors children love

Nutrition Information

CITRUSBERRYTROPICALAPPLE

Nutrient	Per Serving (8 oz)	% Daily Value
Calories	60	3%
Sugar	14g	28%
Vitamin C	45mg	50%
Potassium	200mg	6%

3-column grid for products using CSS Grid

Flexbox for hero section

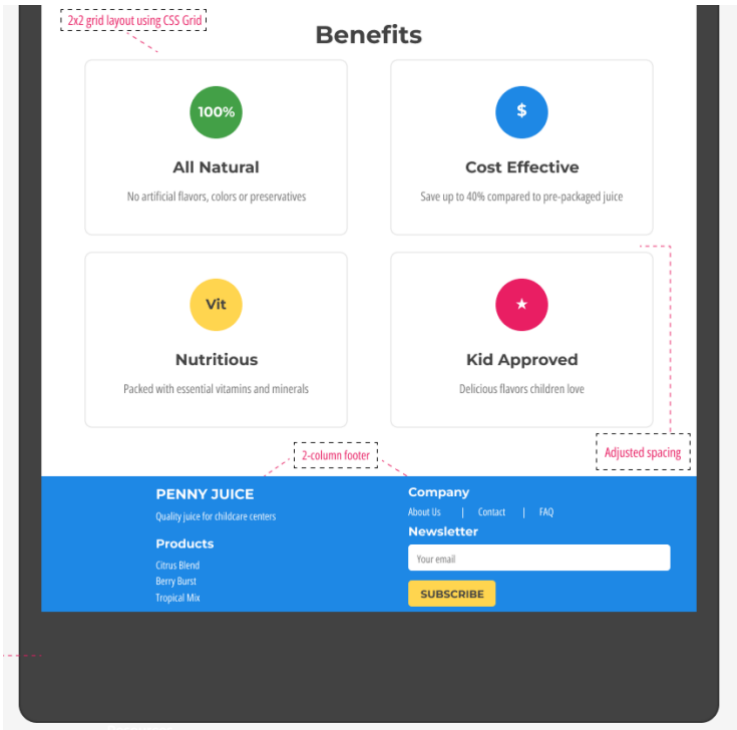
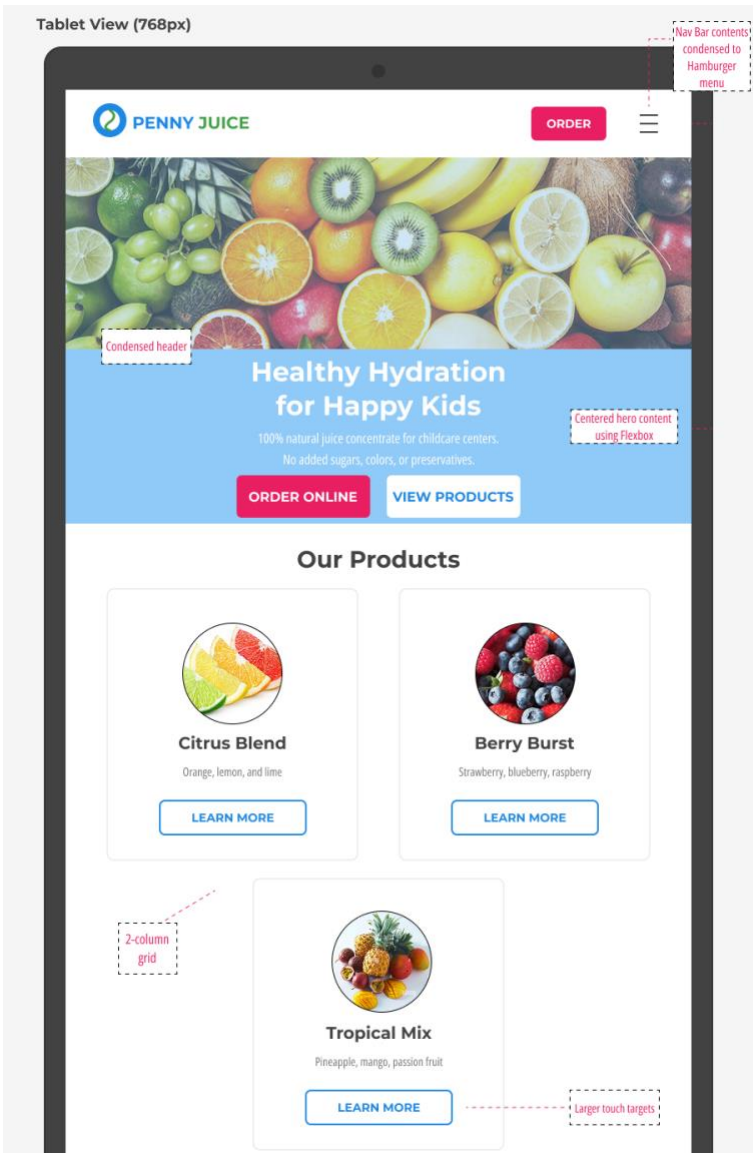
Cards use grid layout with gap

4-column layout for benefits using CSS Grid

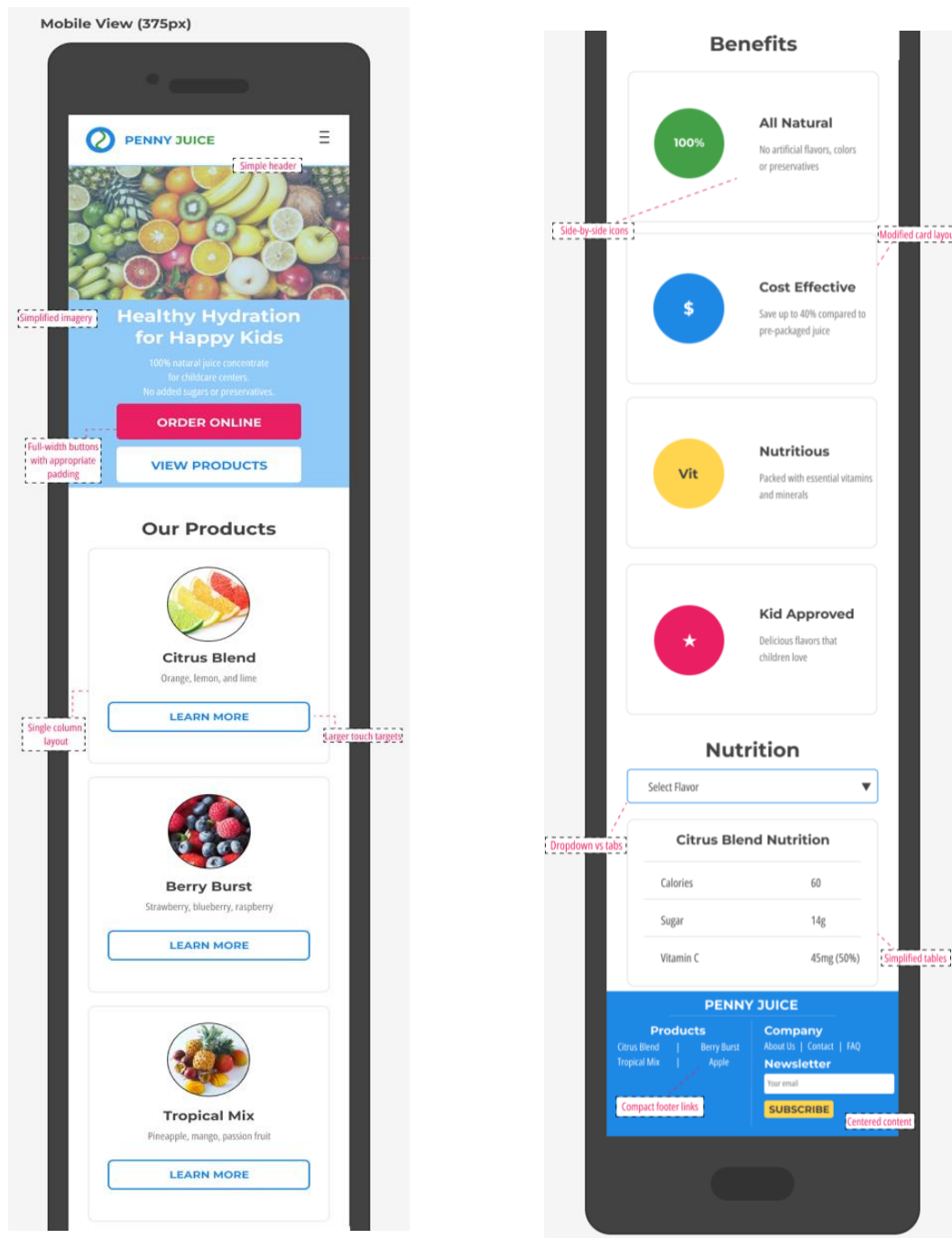
Tabs become dropdown on mobile

Table becomes stacked on mobile

Tablet View Mockup:



Mobile View Mockup:



Additional anecdotes about my design:

- Clear visual hierarchy with logical information flow, consistent headings, and prominent CTAs
- Responsive navigation that adapts across devices while maintaining a consistent mental model
- Clean, professional aesthetic balancing child-friendly elements with B2B presentation
- Enhanced accessibility through high contrast text, proper color usage, and appropriately sized touch targets